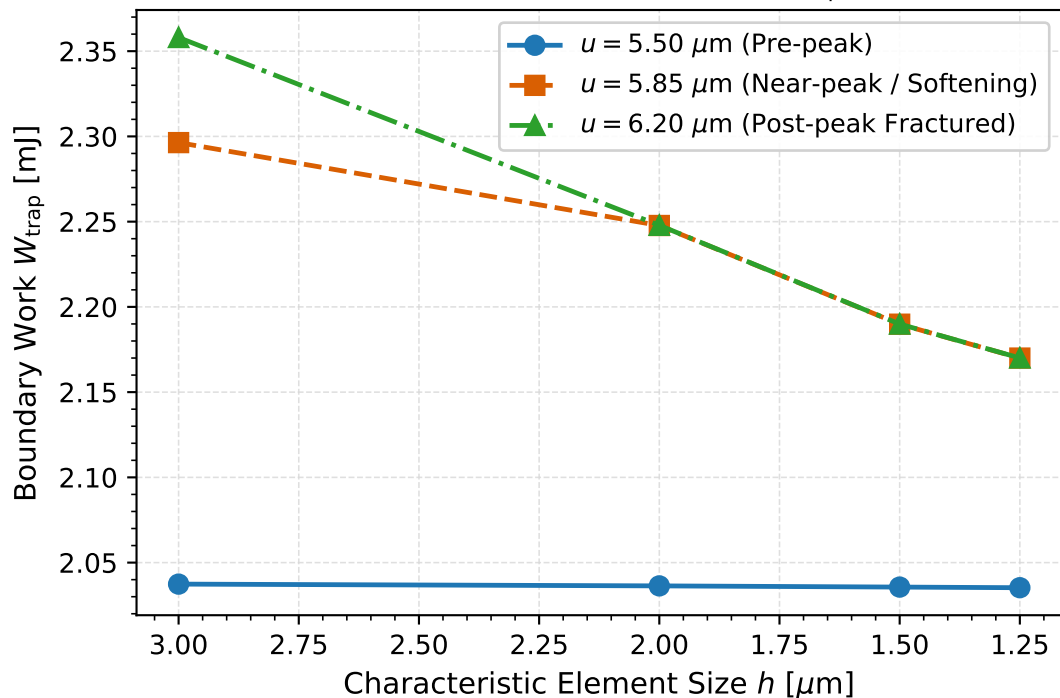
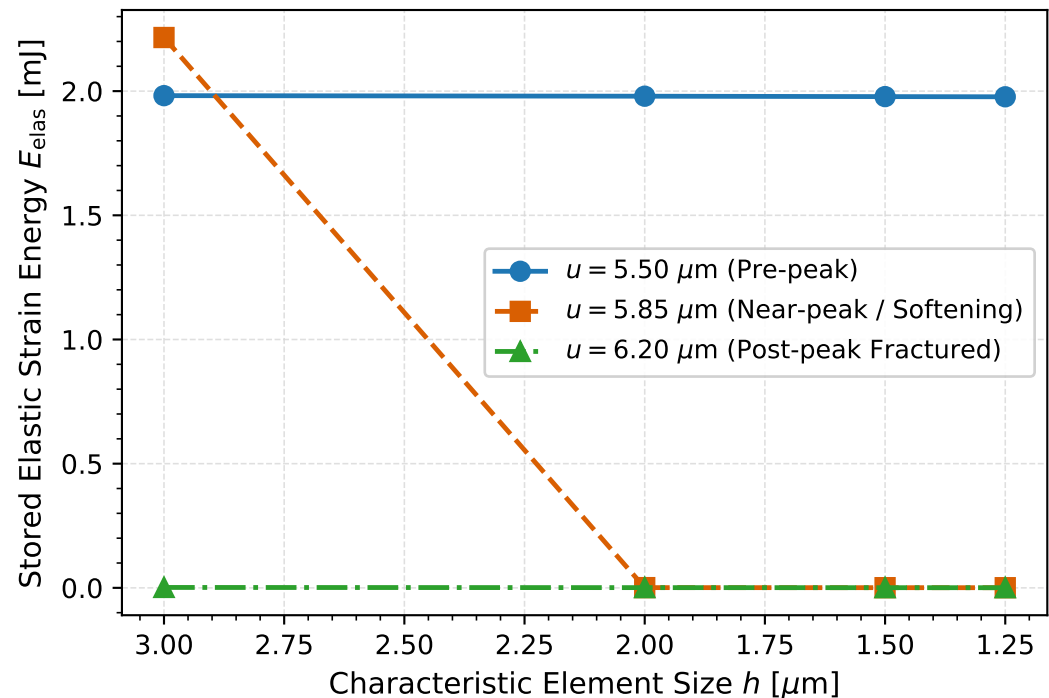


Mode-I Primary Spatial Energy Convergence (S_1 -- S_4 , Single-IP Extraction Rule)

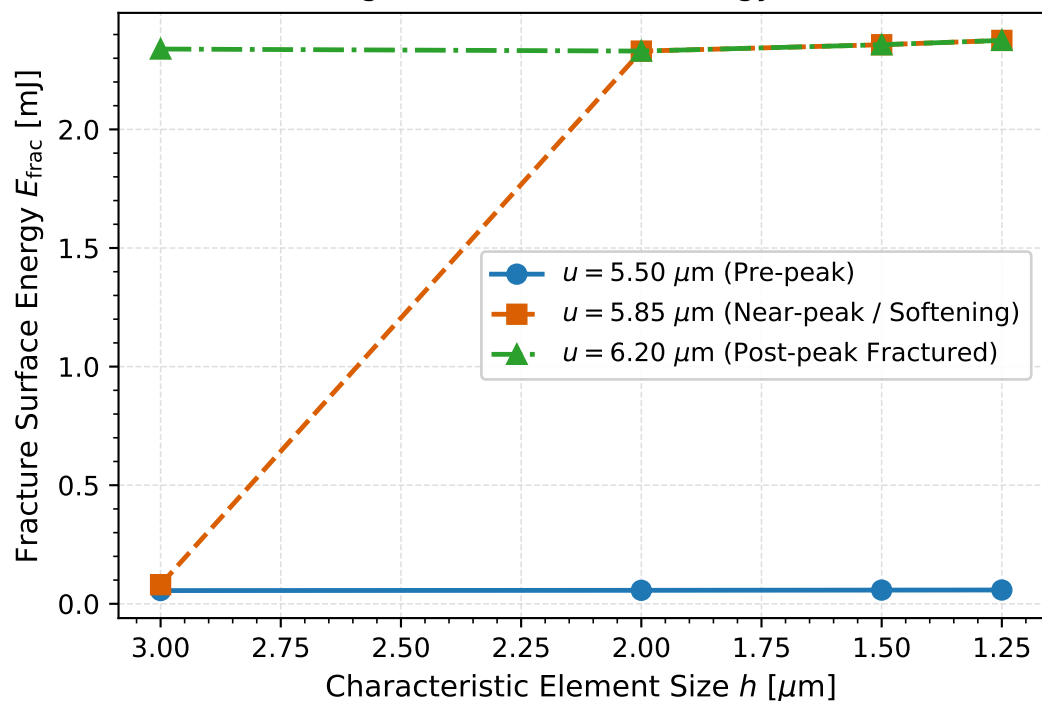
(a) External Boundary Work $W_{\text{trap}}(u)$



(b) Stored Elastic Energy $E_{\text{elas}}(u)$



(c) Regularized Fracture Energy $E_{\text{frac}}(u)$



(d) Two-Term Difference $\Delta_{\text{book}} \equiv W_{\text{trap}} - (E_{\text{elas}} + E_{\text{frac}})$

